

Upscaling high-areal-capacity battery electrodes

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Moving battery technology from the laboratory to large-scale production is a necessary step in achieving cost competitiveness for high-energy-density batteries. So far, academic research has focused on the active material of the electrode and little attention has been paid to cell-level design, hindering the realization of this goal. Therefore, upscaling high-areal-capacity electrode sheets is proposed as a practical way forward. Here we evaluate the impact of high-areal-capacity electrodes on cell energy densities, energy consumption during electrode fabrication and the cost efficiency of cell production. By examining the integration of scalable roll-to-roll electrode-manufacturing techniques (such as slurry casting and dry coating) with the materials chemistry of the electrode components, electrode structure design and cell performance, we aim to outline the areas of development for high-areal-capacity electrodes and provide a structured pathway for bridging the gap between laboratory innovations and industrial scale-up.

The automotive sector is transitioning from internal combustion engines to battery-powered electric vehicles (EVs)¹. The EV market demands high-energy-density batteries ($\geq 350 \text{ Wh kg}_{\text{cell}}^{-1}$) at prices below US\$100 kWh⁻¹, as projected by the United States Advanced Battery Consortium (or USABC) and the US Department of Energy.

Advances across the entire battery value chain^{1,2} include the development of high-capacity/high-voltage active materials³, increased active material content⁴, higher electrode density⁵ and volume-optimized pack efficiency via cell-to-pack configurations. Despite this progress, most academic research has focused on active-material-based approaches to increase their specific energy values⁶ ($\text{Wh kg}_{\text{AM}}^{-1}$), leaving the feasibility of achieving this goal in question. To meet the market targets with the chemistries of both established and emerging active materials, it is imperative for the academic field to consider cell-level strategies that can be universally applied.

Designing high-areal-capacity electrodes by increasing the areal mass loading of active materials offers benefits such

as improved cell energy densities and cost reduction by minimizing inactive components, essential for the widespread adoption of EVs and battery-powered applications. Whereas sophisticated electrode-architecture-driven methodologies have been explored in laboratories⁷, realizing these technologies on an industrial scale requires the identification of challenges from a practical application perspective. The battery industry needs to increase areal capacity without compromising manufacturing scalability and reliability⁸, often overlooked in academic research.

This Review extends beyond academic discussions to address the challenges of upscaling high-areal-capacity electrodes, focusing on roll-to-roll (R2R) slurry-casting/dry-coating manufacturing processes and their effects on electrochemical performance. By integrating development of the electrode component material, optimization of the R2R manufacturing parameters, design of the electrode structure and the charge-transport-kinetics-related performance of the cell, we attempt to facilitate the transition of laboratory-scale innovations in

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high-areal-capacity electrodes into scalable industrial applications. We predict that the electrode-sheet-based strategy will broaden our perspective on enhancing battery performance and cost competitiveness, complementing the conventional approaches for synthesizing novel electrode active materials.

Expanding frontiers in the development of high-energy-density cells

Batteries store and deliver energy via electrochemical reactions, with key components including cathodes, anodes, separators, electrolytes, current collectors and external packaging. A key metric is the specific energy ($\text{Wh kg}_{\text{cell}}^{-1}$), which determines the EV mileage, and is expressed as

$$\text{specific energy of cell} = E_{\text{AM}} \times R_{\text{AM}} \quad (1)$$

Here, R_{AM} , the mass fraction of the cathode active materials in a cell, is defined as

$$R_{\text{AM}} = \frac{M_{\text{AM,cat}}}{M_{\text{cell}}} = \frac{\varphi_{\text{AM,cat}} \times M_{\text{cat}}}{M_{\text{cat}} + M_{\text{an}} + M_{\text{sep}} + M_{\text{elec}} + M_{\text{CC}}}, \quad (2)$$

where $M_{\text{AM,cat}}$ is the mass loading of the cathode active materials and $\varphi_{\text{AM,cat}}$ is the weight fraction of active material in the cathode. In addition, M_{cell} represents the total mass of the cell, which includes the mass of the electrochemically active components (that is, the cathode (M_{cat}) and anode (M_{an})) and the inactive components (that is, separators (M_{sep}), electrolytes (M_{elec}) and current collectors (M_{CC})). The mass of the packaging materials was excluded from this calculation as it varies with cell type. Meanwhile, E_{AM} is the specific energy of an active material, which is calculated via

$$E_{\text{AM}} = V_{\text{AM}} \times Q_{\text{AM}}, \quad (3)$$

where V_{AM} and Q_{AM} denote its nominal voltage and reversible specific capacity, respectively.

Status of prevalent electrode material-based approaches

When the specific energy value of a cell is plotted as a function of the specific capacity value of the cathode active material ($Q_{\text{AM,cat}}$), a direct correlation is shown. In Li-ion batteries (LIBs) based on graphite/Si anodes (Fig. 1a, blue line), an increase in $Q_{\text{AM,cat}}$ from 200 to 250 mAh g^{-1} , and then to 300 mAh g^{-1} , results in energy gains of 14% and an additional 10%, respectively. Despite extensive research on synthesizing advanced cathode active materials over the past few decades, achieving a $Q_{\text{AM,cat}}$ value above 300 mAh g^{-1} remains challenging due to electrochemical instability and cost constraints.

This limitation can be mitigated by exploiting Li-metal battery (LMB) chemistry, which utilizes Li-metal anodes (Fig. 1a, orange line) with the highest theoretical capacity and lowest redox potential². However, with conventional intercalation/deintercalation-based cathodes (LiFePO₄ (LFP), LiCoO₂ (LCO) and LiNi_xMn_yCo_{1-x-y}O₂ (NMC)), which have a $Q_{\text{AM,cat}}$ value typically between 170 and 230 mAh g^{-1} , the specific energy increase of LMBs remains limited. For example, the specific energy values are estimated to be 345 $\text{Wh kg}_{\text{cell}}^{-1}$ at $Q_{\text{AM,cat}} = 200 \text{ mAh g}^{-1}$ and 402 $\text{Wh kg}_{\text{cell}}^{-1}$ even at $Q_{\text{AM,cat}} = 250 \text{ mAh g}^{-1}$. This quantitative analysis indicates that the prevalent active-material-based approaches alone do not sufficiently meet the demand for energy-dense batteries, prompting the need for alternative solutions.

Practical viability of electrode-based approaches

According to equation (1), the specific energy of a cell is influenced by both its E_{AM} (material aspect) and R_{AM} (cell aspect). A higher $M_{\text{AM,cat}}$ increases the numerator relative to the denominator in equation (2), thereby increasing R_{AM} . Given a fixed ratio of negative to positive electrode areal capacity (N/P) and a fixed electrolyte-to-capacity

(E/C) ratio, Fig. 1b shows that as $M_{\text{AM,cat}}$ increases, the proportion of electrodes (blue shading) expands relative to inactive components (orange shading). Figure 1c shows the specific cell energies of LIBs as a function of $M_{\text{AM,cat}}$. At $Q_{\text{AM,cat}} = 200 \text{ mAh g}^{-1}$, increasing $M_{\text{AM,cat}}$ from 15 to 35 mg cm^{-2} raises the specific energy by 14%, which is comparable to the gain from increasing $Q_{\text{AM,cat}}$ to 250 mAh g^{-1} . This estimation is not limited to the chemistry of a particular electrode material. For a material with $Q_{\text{AM,cat}} = 250 \text{ mAh g}^{-1}$, increasing $M_{\text{AM,cat}}$ from 15 to 30 mg cm^{-2} yields a specific cell energy increase that is similar to achieving a practically unattainable $Q_{\text{AM,cat}}$ value of 300 mAh g^{-1} . This demonstrates the viability of the electrode-level approach as an alternative for achieving higher cell energies.

Reducing cost, alongside increasing the specific energy, remains a major challenge in battery technology. This is evident in the EV market's shift from NMC to lower-cost LFP-based cells, despite NMC's higher specific energy. To assess the impact of high-areal-capacity electrodes on cost, we estimated the cost of 100 kWh battery packs as a function of $M_{\text{AM,cat}}$ using the Battery Performance and Cost (BatPaC) model⁹. Increasing $M_{\text{AM,cat}}$ from 15 to 35 mg cm^{-2} reduces the energy consumption in electrode manufacturing by 25% (Fig. 1d). Higher areal-energy-density values of electrodes enable small-scale electrode production while maintaining battery-pack energy requirements, reducing the area required for coating, calendaring and notching. Fewer electrode layers also reduce the number of separators and current collectors, further lowering the material and assembly costs. This simplified cell architecture reduces the cell cost by 14%, bringing it down to US\$110 kWh^{-1} based on the current LIB chemistry, close to the USABC target of US\$100 kWh^{-1} . This analysis underscores the effectiveness of high-areal-capacity electrodes in achieving energy-dense, cost-competitive cells.

Manufacturing scalability of high-areal-capacity electrodes

From a manufacturing perspective, we give a concise overview of previous studies on high-areal-capacity electrodes, categorizing them into two groups: R2R manufacturing (via slurry casting and dry coating) and alternative fabrication techniques. Given the increasing interest in scalable R2R manufacturing, our discussion focuses on key specifications and requirements for the industrial-scale manufacture of electrodes.

Conventional electrode manufacturing relies on slurry-casting technology. This method involves dispersing the electrode components in a processing solvent (for example, *N*-methyl-2-pyrrolidone or water), coating them onto current collectors and then carrying out a drying step. Slurry casting is scalable and offers high throughput due to its compatibility with R2R processing. However, slurry-cast electrodes encounter difficulties in increasing areal capacity due to electrode cracking and redox inhomogeneity.

The dry-coating method, pioneered by Maxwell and Tesla, eliminates the need for processing solvents during the manufacture of electrodes¹⁰. Several techniques have recently been explored, including free-standing film lamination¹¹, electrostatic spray deposition¹², chemical deposition¹³ and melt extrusion. This Review focuses on free-standing film lamination, which is anticipated for potential use in the near-future in battery production lines. This method involves the mixing of dry powder via ball milling, followed by fibrillation with a malleable binder under an applied force. The electrode components, which are entangled by the fibrillated binder, are formed into a free-standing film and calendared onto a current collector. Dry coating eliminates the problematic solvent-drying step that is required by the slurry-casting process¹¹, potentially reducing battery manufacturing costs by approximately 15% and improving the production efficiency of high-areal-capacity electrodes¹⁴.

Meanwhile, innovative electrode fabrication methods such as 3D printing¹⁵, freeze casting¹⁶, templating¹⁷ and phase inversion¹⁸ have been proposed in the laboratory. These techniques aim to fabricate unique

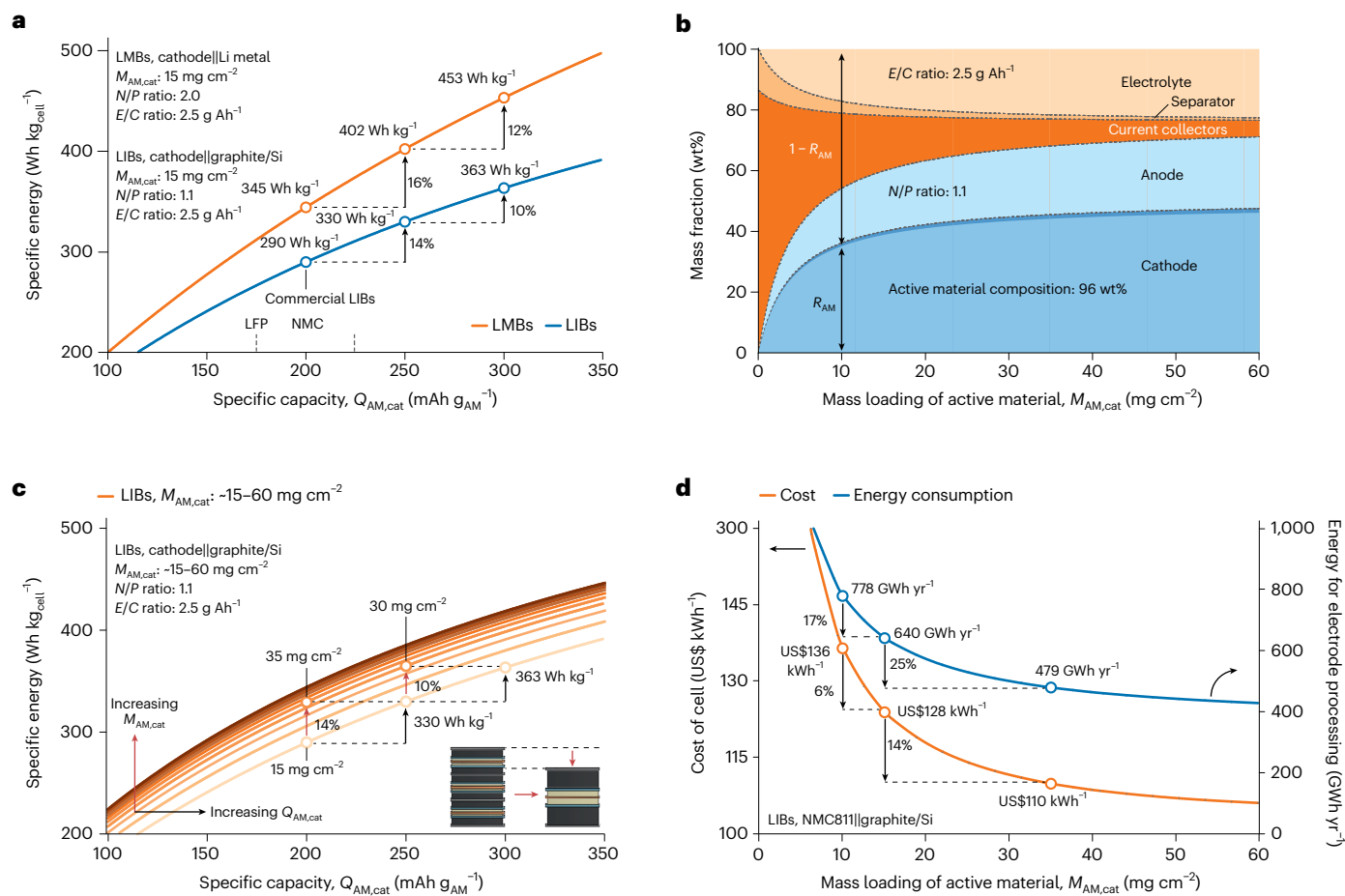


Fig. 1 | Practical viability of electrode-based approaches for achieving high-energy-density batteries. **a**, Specific energy values of LIBs and LMBs as a function of the specific capacity ($Q_{\text{AM,cat}}$) of the cathode material. **b**, Mass fraction of the cell components as a function of the areal mass loading of the active material in the cathode ($M_{\text{AM,cat}}$). The coloured areas represent the contributions from different cell components: cathode (active materials (mid-blue), and binder/conductive additives (dark blue)), anode (lighter blue), current collectors (dark orange), separator (mid-orange) and electrolyte (light orange). **c**, Specific energy values of LIB cells as a function of $M_{\text{AM,cat}}$ (which is varied from ~ 15 to

60 mg cm^{-2}). The inset shows schematics of multi-layer cell configurations where the cells are connected in parallel. Each configuration has the same energy but a different $M_{\text{AM,cat}}$ value. The components are represented as follows: cathodes, dark grey; aluminium current collectors, grey; anodes, stone; copper current collector, orange; and separators, blue. **d**, Cost reduction of cells and energy consumption of electrode manufacturing as a function of $M_{\text{AM,cat}}$. This estimation was conducted using the BatPaC tool⁹ based on a total pack energy of 100 kWh and an annual pack production of $500,000$. NMC811, $\text{LiNi}_{0.8}\text{Mn}_{0.1}\text{Co}_{0.1}\text{O}_2$.

electrode microstructures that mitigate cracking and enhance the electrochemical performance in high-areal-capacity electrodes. However, performance evaluations have focused predominantly on achievable areal capacities or electrode thicknesses, and often neglect factors such as electrode densities or active material contents. This oversight can lead to misleading conclusions about the practical feasibility of these approaches for increasing the energy density at the cell level. Moreover, they often involve complex and time-intensive manufacturing processes, limiting their scalability beyond the laboratory.

The demand for rapid, cost-effective manufacturing requires practical processes that can be easily integrated into existing production lines. This Review focuses on scalable techniques, particularly slurry casting and dry coating, which are compatible with conventional R2R electrode manufacturing. The materials chemistry of the electrode components is also discussed for upscaling high-areal-capacity electrodes.

Challenges of R2R high-areal-capacity electrodes

This section provides an overview of the manufacturing and physico-chemical issues associated with the R2R production of high-areal-capacity electrodes, with a focus on slurry-casting (Fig. 2a) and dry-coating (Fig. 2b) processes.

Manufacturing issues of slurry-cast electrodes

Slurry casting begins with dispersing active materials and conductive additives in powder form, followed by adding binders dissolved in a solvent to form an electrode slurry. Commercial slurry casting uses slot-die coating for high-speed ($>100 \text{ m min}^{-1}$) and scalable applications (coating widths of $>1,500 \text{ mm}$), in addition to its ability to adjust the coating thickness by varying the gap between the die and the current collector. However, increasing the coating thickness causes undesirable thickness variations, enlarging the edge-elevation area¹⁹. This phenomenon is attributed to fluid dynamic imbalances at the slurry-die interface²⁰ (for example, edge stress effect and die swell) and non-uniform solvent drying rates²¹. These inhomogeneities cause uneven electrode loading across the lateral direction. Consequently, regions with a lower loading experience higher depths of charging/discharging, whereas regions with a higher loading suffer excessive stress during calendaring, potentially accelerating degradation²². Furthermore, local discrepancies in the N/P ratios could cause metal plating on the anodes²³.

After coating, solvents are removed via convection drying at elevated temperatures. Drying zones in commercial electrode production lines typically extend up to 100 m in length. Evaporation at the slurry surface drives solvent migration to the top, while diffusion inside the electrode maintains the concentration equilibrium²⁴. The

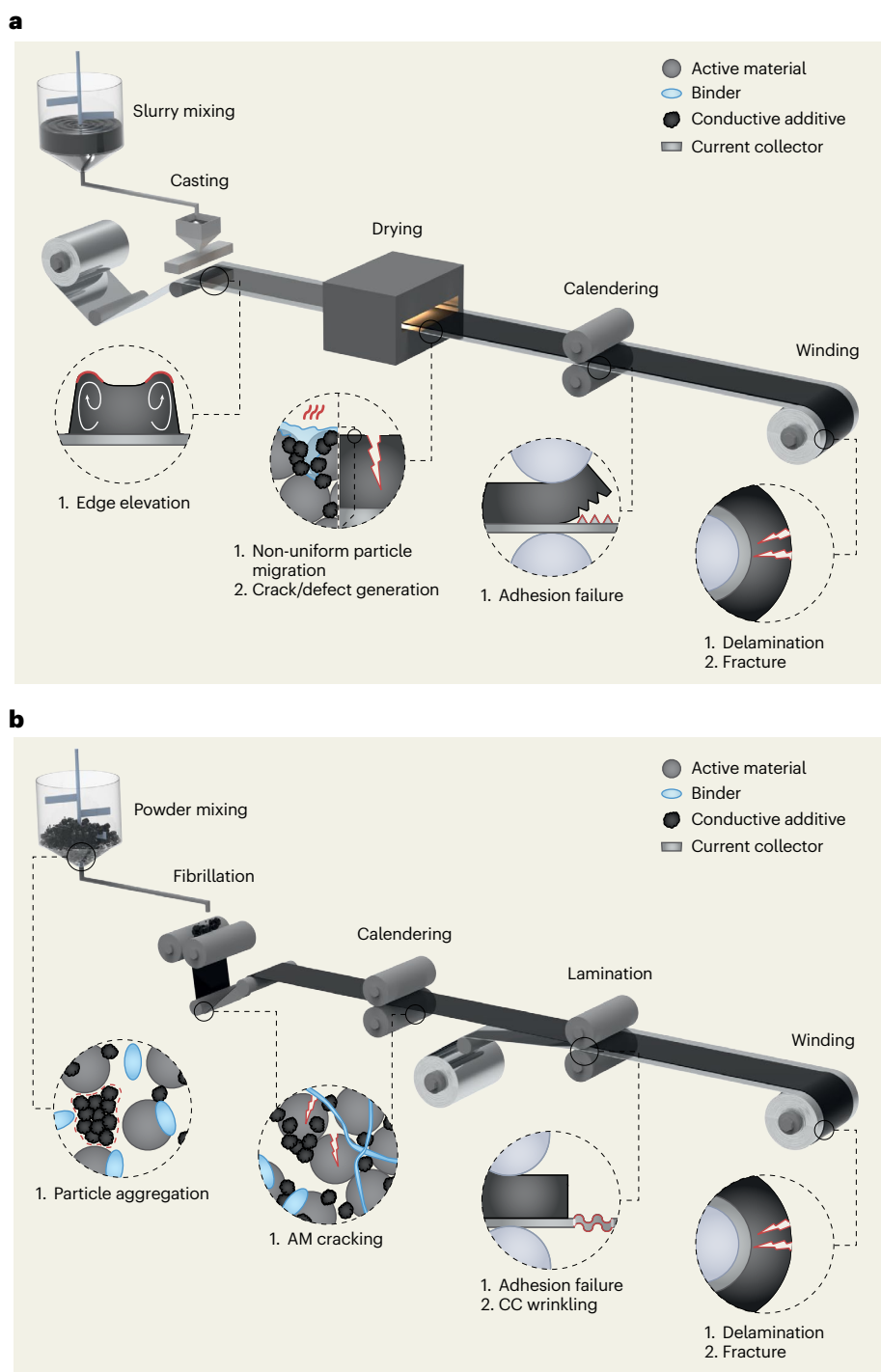


Fig. 2 | Manufacturing and physico-chemical challenges of R2R high-areal-capacity battery electrodes. a,b, Production of high-areal-capacity electrodes via slurry casting (a) and dry coating (b). The expanded insets depict the

potential challenges involved with each step of the process, and include the uneven lateral and cross-sectional distribution of electrode components and structural instability of the electrodes.

balance between diffusion and evaporation is quantified by the Péclet number (Pe)²⁵:

$$Pe = \frac{\dot{E}}{D_0} \times h = \frac{6\pi\mu R\dot{E}}{k_B T} \times h, \quad (4)$$

where D_0 , h , $\dot{E}$, μ , R and $k_B T$ denote the diffusion coefficient, film thickness, evaporation rate, solution viscosity, particle radius and thermal energy, respectively. When $Pe > 1$, evaporation dominates diffusion, causing component accumulation at the top of the film. In thick composite films (high h), when Pe is >1 for both small and large particles,

smaller particles tend to accumulate at the top, especially when the size ratio between the two particles is substantial (for example, size ratio > 3)²⁶. A substantial size difference between the active material (in the range of 1–10 μm) and the carbon conductive additive (approximately 50 nm) causes lightweight conductive additives and dissolved binders to migrate upwards, while heavier active materials settle by gravity²⁷. This segregation impairs electron percolation, particularly in the bottom layer, reducing the through-plane electronic conductivity²⁸. Excessive concentrations of conductive additives and binders in the top layer hinder electrolyte infiltration into the electrodes.

After excess solvent evaporates, solvent menisci form between particles of the electrode. These menisci generate capillary pressure that is directed downwards, inducing tensile stress inside the electrode²⁹. As electrodes comprise rigid materials, this stress is dissipated through film deformation, increasing the susceptibility to cracking above a certain thickness. The critical cracking thickness ($h_{\max}$) is defined as²⁹

$$h_{\max} = 0.41 \left(\frac{GM\phi_{\text{rcp}}R^3}{2\gamma} \right)^{\frac{1}{2}}, \quad (5)$$

where G , M , ϕ_{rcp} , R and γ denote the shear modulus of the particles, coordination number, particle volume fraction at random close packing, particle radius and solvent–air interfacial tension, respectively. For NMC cathodes, $h_{\max}$ is reported to be approximately 170 μm (ref. 30) (for $M_{\text{AM,cat}} = 35 \text{ mg cm}^{-2}$). Although water is a more sustainable and cost-effective solvent³¹, it presents manufacturing challenges for high-areal-capacity electrodes. According to equation (5), the higher surface tension of water (72 mN m^{-1}) compared with *N*-methyl-2-pyrrolidone (40 mN m^{-1}) increases the risk of cracking in thick electrodes.

Once the drying process is complete, roll pressing adjusts the electrode density (2.8–3.4 g cm^{-3} for NMC cathodes and 1.4–1.7 g cm^{-3} for graphite anodes³²) and enhances the surface uniformity. It also improves the electrode cohesion/adhesion, ensuring uniform mechanical stress during cell assembly, including lamination, stacking and winding. However, the modulus disparity between the electrode active layer (in megapascals) and the current collector (in gigapascals) generates shear stress at the interface to maintain strain compatibility during roll pressing³³. In high-areal-capacity electrodes, where binder migration weakens adhesion²⁴, the electrode active layers become more susceptible to delamination from the current collector under the stress.

During final manufacturing (slitting, vacuum drying, storage and transferring to cells), wound electrodes experience bending stress. Electrodes close to the centre of curvature face compressive stress, whereas those further away experience tensile stress³⁴. The strain distribution relative to the distance from the neutral plane (where the net stress is zero) is given by³⁵

$$\epsilon_x = k \times (h_n - y), \quad (6)$$

where k is the bending curvature, which is inversely proportional to the bending radius. In addition, h_n and y are distances from the pivot centre to the neutral plane and a specific location, respectively. Owing to the disparity in Young's modulus between the electrode and the metallic current collector, the neutral plane is located near the current collector³⁵. As the electrode thickness is increased, the distance from the neutral plane to the surface ($h_n - y$) grows, leading to a higher vertical strain (ϵ_x). This strain triggers the formation of convex and concave cracks in double-sided high-areal-capacity electrodes³⁴, particularly at the inner windings near the mandrel of cylindrical cells and the curved parts of flat-wound prismatic cells³⁶. This imposes constraints on the cell design and internal diameters for high-areal-capacity electrodes.

Manufacturing issues of dry-coated electrodes

The dry-coating process, which eliminates the need for processing solvents, prevents the issues associated with the coating and drying of slurry-cast high-areal-capacity electrodes³⁷. However, achieving a uniform dispersion of electrode components remains challenging³⁸. On the basis of the surface energy (γ) values of the electrode components, the work of adhesion (cohesion) between heterogeneous (homogeneous) materials can be defined using the Fowkes equation³⁹:

$$W_{ij} = 2 \left(\gamma_i^d \times \gamma_j^d \right)^{0.5} + 2 \left(\gamma_i^p \times \gamma_j^p \right)^{0.5}, \quad (7)$$

where γ_i^d and γ_j^d represent the dispersive surface energy values of material i and j , respectively, γ_i^p and γ_j^p denote the polar surface energy values of these materials¹². This relationship indicates that disparities in the surface energies of the electrode materials affect their dispersion. Carbon conductive additives, carbon nanotubes in particular⁴⁰, tend to cohesively agglomerate themselves, hindering the formation of uniform electric conductive networks and triggering uneven electrochemical reactions across the electrode.

During dry coating, binders play a pivotal role in maintaining the structural integrity of the electrodes. Polytetrafluoroethylene (PTFE) is commonly used for its ability to fibrillate under shear forces; it physically entangles electrode materials and enables the production of free-standing electrode sheets⁴¹. PTFE fibrillation occurs due to its hexagonal crystal structure, where van der Waals repulsive forces from electronegative of fluorine atoms enable dislocation slip along the polymer chains⁴². However, the non-polar nature of PTFE hinders chemical bonding with the current collectors, resulting in poor adhesion. To address this, a high pressure is applied during calendaring and lamination to mechanically interlock the electrode and the current collector. Such pressure adversely imposes considerable stresses on both the active materials and current collectors. In polycrystalline NMC cathodes, which comprise small primary particles, high compressive stress induces intergranular cracking⁴³, exposing more surface area to the electrolyte and intensifying side reactions⁴⁴. Given the lean electrolyte conditions in practical cells⁴⁵, such electrolyte consumption in thick electrodes induces loss of the ionic transport path during cycling. The applied stress can also cause wrinkling at the uncoated edges of current collectors⁴⁶, which is particularly evident with softer aluminium compared with copper foils²³, potentially tearing the electrodes and hindering intimate contact between electrode layers during cell assembly. The challenges in the dry-coated electrodes during the winding process are similar to those observed in the slurry-cast electrodes. These underscore the need for the careful design of material properties and manufacturing techniques during the R2R production of high-areal-capacity electrodes.

Electrochemical issues of high-areal-capacity electrodes

The charging/discharging of batteries involves complex types of electrochemical resistance, including electron transport through the current collector/electrode (R_e), ion transport in the bulk electrolyte (R_{ion}), ion transport through electrode interstitial pores (R_{ion}), charge transfer at the surface of the active material (R_{ct}) and solid-state diffusion inside the active material. In thin electrodes, where the transport paths of electrons and ions are relatively short, R_{ct} and solid-state diffusion dominate the overall cell performance. However, as the electrode thickness is increased, the longer transport paths of electrons and ions cause R_e and R_{ion} to increase, playing a more significant role in limiting the performance⁴⁷. This disparity arises from a change in the geometry of each type of resistance with the electrode thickness⁴⁸.

The transport phenomena of electrons and ions in a thick cathode during lithiation are illustrated in Fig. 3a. Electrons, generated via oxidation at the anode, move through external circuits and the current collector, with their concentration diminishing towards the top layer of the cathode. In slurry-cast high-areal-capacity electrodes, the upward migration of conductive additives during solvent drying leads to the poor percolation of electrons in the bottom region, increasing the through-thickness electronic resistance⁴⁹. Consequently, electrochemical reactions occur preferentially in the electron-rich regions near the current collector, while the electron-deficient top regions remain inactive²⁸. In dry-coated electrodes, an uneven distribution of the conductive additive further exacerbates the heterogeneity of electrochemical reactions⁵⁰.

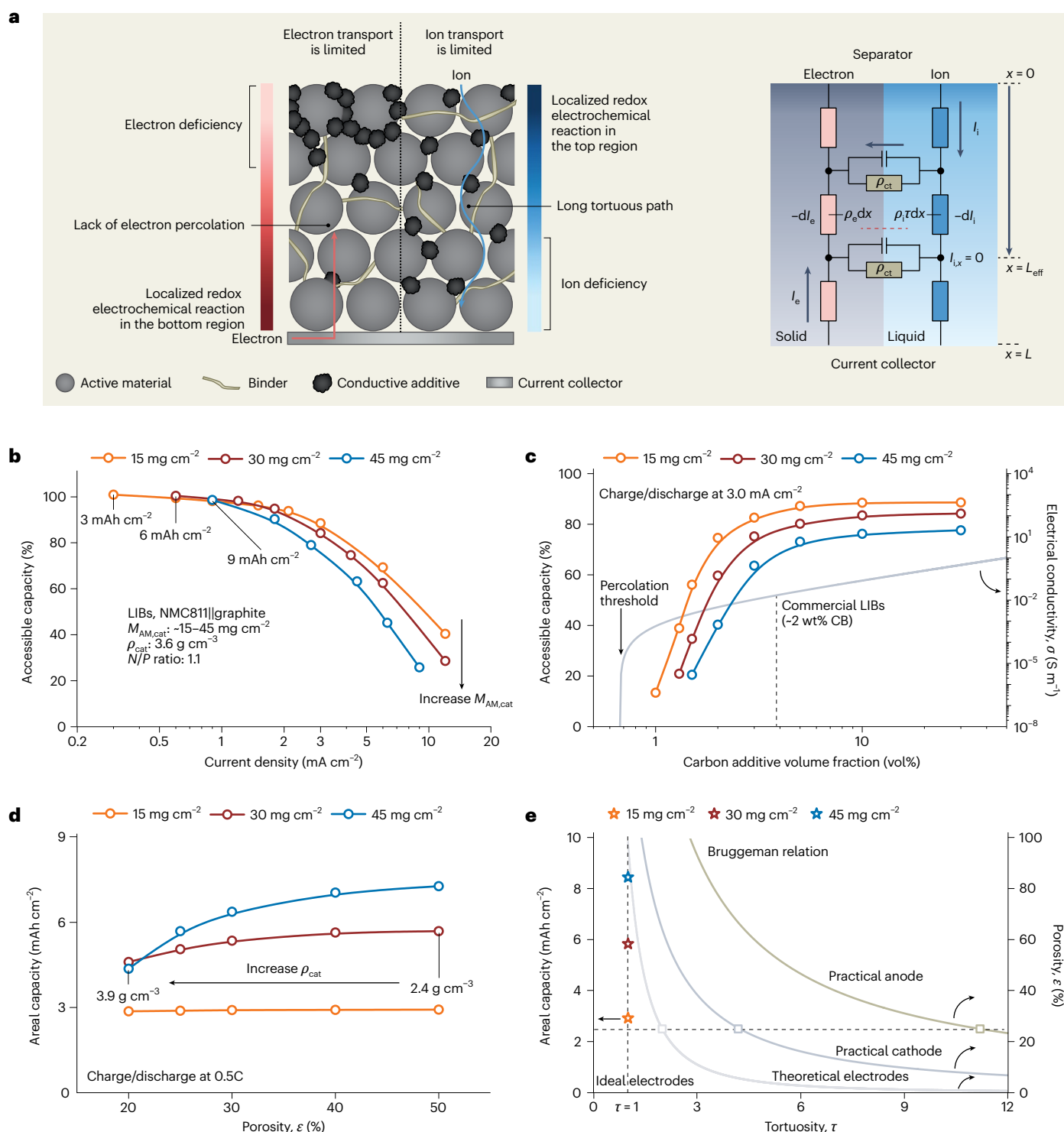


Fig. 3 | Electrochemical challenges of the high-areal-capacity battery electrodes. **a**, Left: schematic illustration of the non-uniformity of an electrochemical reaction in the through-thickness direction of a high-areal-capacity electrode considering electron and ion transport. Right: the effective electrode thickness (L_{eff}) is described as a simplified equivalent circuit on the basis of transmission line model theory⁵⁶. I_i and I_e represent the ionic and

electronic currents, respectively, while ρ_i , ρ_e and ρ_{ct} denote the resistivities of ionic transport, electronic transport, and interfacial charge transfer, respectively. **b–e**, Influence of electrode parameters on the electrochemical metrics simulated based on COMSOL simulations: $M_{\text{AM,cat}}$ (**b**), carbon content (**c**), electrode porosity (**d**) and tortuosity (**e**).

Ion transport typically involves longer distances and higher characteristic time constants than electron transport⁴⁸. It is necessary to consider the effective diffusivity of ions (D_{eff}) in an electrode, as it deviates from the properties of bulk electrolytes due to the tortuous path of ions in porous electrodes. D_{eff} is expressed as⁵¹

$$D_{\text{eff}} = \frac{\varepsilon}{\tau} \times D, \quad (8)$$

where ε , τ and D denote the electrode porosity, tortuosity and electrolyte diffusivity, respectively. A tortuosity of $\tau = 1$ indicates a straight

transport path. The Bruggeman relation defines the relationship between tortuosity and porosity as⁵²

$$\tau = \gamma \times \varepsilon^{1-\alpha}, \quad (9)$$

where γ and α are a scaling factor and the Bruggeman exponent, respectively. The parameter γ is typically in the range of 1–2, whereas α is in the range of 1.5–1.7 in commercial electrodes, resulting in tortuosity values of 2.5–5 for the electrodes⁵³. This shows that ions travel a longer path (τdx) than electrons (dx) in electrodes. Consequently, the ion concentration gradient through the thickness inhibits electrochemical reactions, particularly in the bottom region, while inducing overcharging in the top region of high-areal-capacity electrodes. This is considered to be a major kinetic limitation over electron-related issues⁵⁴, particularly in Li and Na battery electrodes with organic electrolytes as opposed to Zn battery electrodes with aqueous electrolytes. Such transport behaviour is illustrated using an equivalent circuit model^{55,56} in Fig. 3a (right).

The ion-transport behaviour has been theoretically elucidated using the Newman equation⁵⁷ and the derived penetration depth model⁵⁸, which quantifies capacity loss due to ion-transport limitations. The effective penetration depth for electrolyte transport (L_{eff}) is expressed as⁵⁸

$$L_{\text{eff}} = \frac{\varepsilon}{\tau} \times \frac{D_0 C_0 F}{(1 - t_+) I}, \quad (10)$$

where D_0 , C_0 , t_+ , F and I denote the electrolyte diffusion coefficient, electrolyte concentration, cation transference number, Faraday constant and current density, respectively. Assuming that neither solid-state diffusion within the active material nor electronic conductivity are limiting factors, the effective capacity of the electrode (Q_{eff}) relative to its theoretical capacity (Q_0) is formulated as⁵⁹

$$\frac{Q_{\text{eff}}}{Q_0} = \frac{L_{\text{eff}}}{L} \quad (11)$$

$$Q_{\text{eff}} = Q_0 \times \frac{\varepsilon}{L\tau} \times \frac{D_0 C_0 F}{(1 - t_+) I} \quad (12)$$

As the electrode thickness (L) exceeds L_{eff} , active materials in the bottom region struggle to participate in the electrochemical reaction due to ion deficiency, thereby reducing Q_{eff} . This causes the upper regions, which are relatively ion-rich, to experience higher states of charge and depths of discharge, thus accelerating degradation and electrolyte decomposition⁵⁴. In addition, the accumulation of by-products further impedes ion transport, exacerbating electrochemical inhomogeneity and degrading the long-term cycling performance⁶⁰. This problem becomes more evident as the applied current density (I) is increased, underscoring the limitations of high-areal-capacity electrodes in terms of power density and fast charging capabilities. For EV batteries that are suitable for long-range driving and fast charging, comparable to internal combustion engines, managing effective charge-transport distances in high-areal-capacity electrodes is a paramount concern. Equation (10) provides key insights for maximizing L_{eff} , considering both electrode (ε , τ and t_+) and electrolyte (C_0 , D_0 and t_+) parameters.

To quantify such deliverable capacities under various conditions, we used COMSOL simulations based on the theoretical model framework established by Newman and co-workers⁵⁷, incorporating electrode parameters with $M_{\text{AM,cat}}$ in the range of 15–45 mg cm⁻². Pseudo-two-dimensional models of the NMC811||graphite cells with cathode areal capacities of 3.0, 6.0 and 9.0 mAh cm⁻², corresponding to the respective $M_{\text{AM,cat}}$ values of 15, 30 and 45 mg cm⁻², were constructed using the electrolyte of 1 M LiPF₆ in ethylene carbonate/diethyl carbonate. The tortuosity and porosity of the cathodes and anodes

follow the Bruggeman relationship, as introduced in equation (9). The dependence of the rate capability on $M_{\text{AM,cat}}$ was investigated in NMC811||graphite cells, assuming a commercial-level cathode density (ρ_{cat}) of 3.6 g cm⁻³ (corresponding to a porosity of 25%), within a voltage range of 2.8–4.3 V. As expressed in equation (12), Q_{eff} is inversely proportional to both L and I , indicating that thicker electrodes or higher current densities reduce active material utilization. In Fig. 3b, at a low current density (1 mA cm⁻²), accessible capacities close to 100% were achieved across all $M_{\text{AM,cat}}$, in which case high-areal-capacity electrodes exhibit their merits on specific energies, as shown in Fig. 1c. At a practical current density (3 mA cm⁻²), 76% of the maximum capacity was retained for the cathode with $M_{\text{AM,cat}} = 45$ mg cm⁻², 7% and 12% lower than the cathodes with $M_{\text{AM,cat}}$ values of 30 and 15 mg cm⁻², respectively. The differences in the accessible capacity between these cells become more pronounced at higher current densities. For instance, at a current density of 10 mA cm⁻², premature discharge termination is mainly due to Li⁺ concentration polarization in high-areal-capacity electrodes, which lowers the discharge plateau and accessible capacity. Active materials near the separator remain readily accessible to Li⁺, whereas those near the current collector face tortuous ion-transport limitations. This result highlights a trade-off between specific energy and power in high-areal-capacity electrodes.

Conductive additives are essential in forming electrical conduction networks and providing efficient electron transport, particularly in electrodes with low-conductivity active materials (for example, LFP). Carbon black (CB) is commonly used in commercial electrodes, with a minimum content of 2 wt% (or 3.8 vol%) for electrical percolation. In thicker electrodes, electron transport across the electrode's thickness requires a higher through-plane electrical conductivity (σ_{TP}) to allow electrons to efficiently travel between the top and bottom layers within limited charge/discharge time. On the basis of the electrical percolation behaviour studied in NMC811–CB composites⁶¹, we observed a strong correlation between the accessible capacity and additive volume fraction in electrodes with various $M_{\text{AM,cat}}$ values (Fig. 3c). As $M_{\text{AM,cat}}$ is increased, the minimum fraction of conductive additive required to maintain accessible capacity also increases. This suggests that higher amounts of conductive additives, or those with a lower percolation threshold, are required for efficient electron transport in high-areal-capacity electrodes.

The spatial pore geometry of electrodes, with key structural parameters (ε and τ), governs the ion-transport kinetics inside the electrolyte-filled pores. According to equations (9) and (10), higher ε and lower τ promote ion transport, increasing L_{eff} . In Fig. 3d, increasing ε from 20 to 50% resulted in a higher capacity increase in the cathode with $M_{\text{AM,cat}} = 45$ mg cm⁻² than it did for the cathodes with $M_{\text{AM,cat}} = 30$ or 15 mg cm⁻². This difference in performance is attributed to the effectiveness of the higher ε in reducing problematic concentration gradients, particularly in high-areal-capacity electrodes, thereby improving the overall performance. Although creating more pores seems to be an effective strategy for high rate capabilities, it decreases the electrode density (for example, $\rho_{\text{cat}} = 2.4$ g cm⁻³ when $\varepsilon = 50\%$) and cell energy density, countervailing the benefits of high-areal-capacity electrodes.

A plausible strategy is to decouple ε and τ , eliminating their conformity to the Bruggeman relationship. In practical battery electrodes, the measured τ is often larger than the theoretical values, primarily due to the carbon binder domain and anisotropic geometry of the active materials (Fig. 3e). For non-spherical particles such as graphite, their packing during electrode preparation increases τ in the through-thickness direction⁶². Creating vertically aligned channels for ion transport could decrease τ close to unity while maintaining a low ε . With an ideal value of $\tau = 1$, areal capacities can still be retained in the cathode with $M_{\text{AM,cat}} = 45$ mg cm⁻² and $\rho_{\text{cat}} = 3.6$ g cm⁻³, indicating that tortuosity regulation is a promising strategy for addressing the electrochemical inhomogeneity and achieving high energy/power in high-areal-capacity electrodes.

Table 1 | Materials chemistry of scalable high-areal-capacity electrodes

Electrode materials	Issues	Underlying causes	Material properties	Proposed solutions
Active materials	Particle cracking (manufacturing)	High calendering pressure	Elastic modulus	(1) Grain boundary engineering for increased intraparticle cohesion (2) Concentration gradient design for increased reversibility under stress (3) Single-crystalline materials for elimination of grain boundaries
			Hardness	
			Fracture strength	
	Ion deficiency (electrochemical)	Effective electrode length	Particle size	(1) Large particle size for reduced tortuosity (2) Highly conductive materials for lower carbon content and tortuosity (3) Spherical particles for reducing the anisotropic tortuosity
			Conductivity	
			Geometry	
Binders	Electrode crack (manufacturing)	Drying-triggered tensile stress	Toughness	High-toughness and mechanically resilient polymers
	Delamination, fracture (manufacturing)	Vertical strain	Tensile properties	Flexible and adhesive polymers that are resistant to concave/convex fracture
	Electron deficiency (electrochemical)	Particle aggregation	Surface energy	Surface energy matching with heterogeneous materials for dispersion
	Ion deficiency (electrochemical)	Effective tortuous path of electrolytes	Polarity	Enhanced binder–electrolyte interaction for improved electrolyte wetting while suppressing swelling
Conductive additives	Electron deficiency (electrochemical)	Particle aggregation	Surface energy	(1) Surface energy matching with heterogeneous materials for dispersion (2) Higher aspect ratio that can form conductive networks at low contents
		Particle migration	Aspect ratio	
Current collectors	Delamination (manufacturing)	Surface energy mismatch	Roughness	(1) Etching for increased surface area (2) Primer layers for controlled surface energy and interfacial affinity
	Chemical bond			
	Heat accumulation (electrochemical)	High internal resistance	Thermal conductivity	(1) Higher thermal conductivity for heat dissipation (2) Flame-retardant properties for suppressed thermal runaway

Materials chemistry of scalable high-areal-capacity electrodes

The types of materials chemistry that affect the manufacturing scalability and electrochemical reliability of high-areal-capacity electrodes are discussed below, with a focus on the key issues, underlying causes, material properties and proposed solutions (Table 1).

Active materials

The high pressure applied during calendaring and lamination often causes cracks in active materials. Therefore, structurally durable active materials with sufficient elastic modulus, hardness and interfacial fracture strength are preferred⁶³. For polycrystalline NMC materials, strategies such as grain boundary engineering⁶⁴ and concentration gradient design⁶⁵ have been developed to improve mechanical durability. The transition to single-crystalline structures is another promising solution as it eliminates grain boundaries that are prone to cracking⁶⁶. However, the scalable production of single-crystalline materials requires precise control of deagglomeration, the temperature gradients over the large batches involved and the calcination time, increasing the synthesis complexity and cost⁶⁷.

From the perspective of the electrochemical performance, active materials with a low specific surface area and high electronic conductivity are preferred. Whereas small particles offer shorter solid-state diffusion paths and rapid charge transfer due to their larger surface area, they also have increased tortuosity⁶⁸ and require more conductive additives for percolation, hindering mass transport in electrodes⁶⁹. Compared with layered transition metal oxides, polyanionic olivine materials such as LFP and $\text{LiMn}_x\text{Fe}_{1-x}\text{PO}_4$ (LMFP) typically utilize smaller particles to compensate their low solid-state diffusivity, making their use in thicker electrodes challenging. Ion doping⁷⁰ and carbon coating⁵ can reduce reliance on conductive additive content, thereby

decreasing the ion-transport tortuosity⁷¹. Another critical factor is particle geometry. Graphite, which is used widely as an anode in LIBs, has an anisotropic, flake-like structure. It increases the tortuosity and concentration gradients in thicker anodes⁷². Li-ion deficiency increases the likelihood of metallic Li deposition on the graphite, which consumes reversible Li and raises the risk of internal short circuits and safety issues⁷³. Reshaping graphite into spherical particles has been explored⁷⁴. Blending high-capacity Si-based anode materials, such as Si–C composites and SiO_x , with graphite in optimized ratios can reduce the electrode thickness (that is, the charge-transport distance) to achieve the desired N/P ratio without compromising the structural stability⁷⁵.

Binders

Binders ensure electrode cohesion and strong adhesion to current collectors, maintaining structural integrity during manufacturing and operation. The physical properties of binders must prevent problems such as electrode cracking during drying and delamination during winding. A key factor is binder toughness—the ability to absorb energy before failure—which helps to withstand drying-induced tensile stress. High-toughness electrodes mitigate cracking during solvent drying, enhancing the critical cracking thickness^{4,30}. Describing the seamless contact that must be maintained between the electrode active layer and the current collector during winding can be expressed via⁷⁶:

$$f = \frac{1}{r_b} = \left(\frac{2}{h}\right) \times \left(\frac{\sigma_y}{E}\right). \quad (13)$$

Flexibility (f) is quantified by the minimum bending radius before failure (r_b), yield stress (σ_y) and Young's modulus (E), along with the electrode thickness (h). As active materials are rigid, the binder determines

the electrode's elastic properties. Consequently, binders with a high yield stress and a low Young's modulus are ideal for high-areal-capacity electrodes in R2R production. These properties can be enhanced by blending different binders with high toughness and flexibility via interpenetrating networks⁴ or chemical bonding⁷⁷.

Binders can interact with heterogeneous materials. Interactions between binders and conductive additives are governed by the surface energy parameters^{12,38}. Binders with unsaturated aromatic groups⁷⁸ enhance interaction with conductive additives via π - π stacking, promoting uniform dispersion. Binders also interact with electrolytes, establishing ion-percolation channels. Polar groups in binders improve electrolyte infiltration⁷⁹, but the binders can easily swell, weakening adhesion between the electrode active layer and current collector. Recent studies have shown that binders influence ion transport by regulating electrolyte coordination in the electrode pores⁴. These findings suggest reconsidering the role of binders beyond their traditional function as adhesives.

Conductive additives

Conductive additives provide electron-conduction pathways, where their morphology affects both the electrical conductivity and mechanical strength of electrodes. Electron-conduction networks in composites follow percolation threshold theory, which states that above a critical volume fraction of the conductive filler, the conductivity of the composite approaches that of the filler⁸⁰:

$$\sigma = \sigma_0 \times \left(\frac{\varphi - \varphi_c}{1 - \varphi_c} \right)^t, \quad (14)$$

where σ and σ_0 represent the electrical conductivity values of the composite and filler, respectively, and φ , φ_c and t denote the filler volume fraction, critical volume fraction and the percolation exponent, respectively.

The electrical conductivity can be increased either by increasing the additive fraction, which is detrimental to specific cell energies, or using additives with low percolation thresholds. Conductive additives with higher aspect ratios, such as one-dimensional (1D) materials including single-walled carbon nanotubes ($\varphi_c = 0.16$ vol%) and multi-walled carbon nanotubes, are easier to form interconnected networks, thereby reducing the percolation threshold⁸¹. These 1D additives exhibit higher conductivity than zero-dimensional additives such as CB ($\varphi_c = 0.67$ vol%), even at lower fractions⁶¹. The relationship between aspect ratio and conductivity has been experimentally validated, demonstrating that electrodes incorporating 1D conductive additives exhibit higher through-thickness conductivity and improved rate capability in high-areal-capacity electrodes⁸². In addition, the higher aspect ratio of 1D additives entangles the electrode components, improving the tensile and yield strength of electrodes^{37,83}, which is beneficial for structural stability during R2R manufacturing. However, the 1D conductive additives tend to form large agglomerates, making uniform dispersion challenging. Surface energy modification of the conductive additives has been proposed, either to increase interparticle repulsion or to enhance their affinity with active materials¹².

Current collectors

Current collectors transport electrons and provide a uniform electron distribution across large electrode areas and connection to external circuits. Their surface properties affect adhesion to electrode slurries or electrode films, impacting the manufacturing yield and performance of electrodes⁸⁴. For dry coating, adhesion is achieved via mechanical interlocking, whereby the electrode film embeds into the surface of the current collector. However, the rigid, flat surfaces of commercial current collectors restrict contact with the electrode active layers. The contact area can be increased via chemical etching⁸⁵ or by applying primer layers of low-modulus materials, such as carbon or polymer⁸⁶.

Whereas these modifications improve the production yields and cell performance of dry electrodes, they may reduce the cell energy densities and increase costs.

High-areal-capacity electrodes generate more heat due to their higher resistance. To quantify this, we calculated the heat generation rate and temperature changes in a 100 kWh pack using the BatPaC tool⁹. Increasing the electrode loading from 15 to 35 mg cm⁻² doubles the heat generation, raising the cell temperature from 34 to 40 °C at the end of discharge in an adiabatic system. Current collectors serve as thermal conductors, managing heat accumulation and thermal runaway. Incorporating flame-retardant materials into copper-coated polyimide films⁸⁷ or thermally responsive layers⁸⁸ that inhibit electrochemical reactions above certain temperatures can enhance the thermal safety. Materials with high thermal conductivity and stability, such as graphene⁸⁹, dissipate heat more effectively than conventional metallic current collectors, reducing local heat-spots and preventing thermal runaway.

Discussion

This Review highlights the importance of collaborative efforts to realize high-areal-capacity electrodes via scalable R2R manufacturing techniques, with a focus on slurry casting and dry coating. Whereas the initial development of battery technology typically begins at the laboratory scale, where material and electrode properties are investigated using small-scale laboratory equipment, these early-stage studies often overlook the complexities that are inherent to large-scale production and application-specific challenges. Integrating an understanding of industrial processes into academic research is key to advancing technology readiness levels (TRLs)⁹⁰.

Scaling strategies for high-areal-capacity electrodes

For upscaling high-areal-capacity electrodes, we have outlined the areas of development regarding the materials chemistry of the electrode components, R2R manufacturing, electrode structure design and cell performance (Fig. 4). Beyond the performance improvements, considerations of elemental abundance, material cost and synthesis maturity of the electrode components are imperative for ensuring practical applications.

The manufacturing parameters in particle mixing (for dry-coated electrodes), slurry mixing/drying (for slurry-cast electrodes), calendering and winding should be optimized to ensure the structural stability and productivity of high-areal-capacity electrodes. For dry-coated electrodes, achieving uniform particle mixing within a limited time frame is crucial, considering the optimized degree of PTFE fibrillation⁹¹. For slurry-cast electrodes, binder migration and crack formation during drying should be considered to avoid structural defects in subsequent processes. The preparation of well-dispersed, high-solid-content slurries suppresses binder migration and sedimentation of the active material due to increased viscosity⁹². Multi-stage or infrared drying can provide a uniform distribution without compromising the drying rate⁹³. During calendering, adjusting the roll pressure (p_{roll}) or feed speed (v_{feed}) can reduce line load and shear stress, thereby preventing delamination. Optimizing yield and throughput according to the electrode properties is required. For winding processes (or cylindrical cells), reels (or mandrels) with a smaller radius improve the packing efficiency and enable variations in the cell dimensions, although they also increase convex/concave stress, heightening the risk of cracking.

Active material compositions should exceed 96 wt% while maintaining the structural integrity and electronic conductivity. Recently, the battery industry has implemented multi-layer coating techniques (for example, simultaneous slot-die coating) at both pilot and commercial scales. This approach addresses issues such as inhomogeneous component distribution and degradation of the active material during calendering by controlling the particle type, composition⁹⁴ and porosity⁹⁵ of each layer. For example, smaller active materials in the top layer

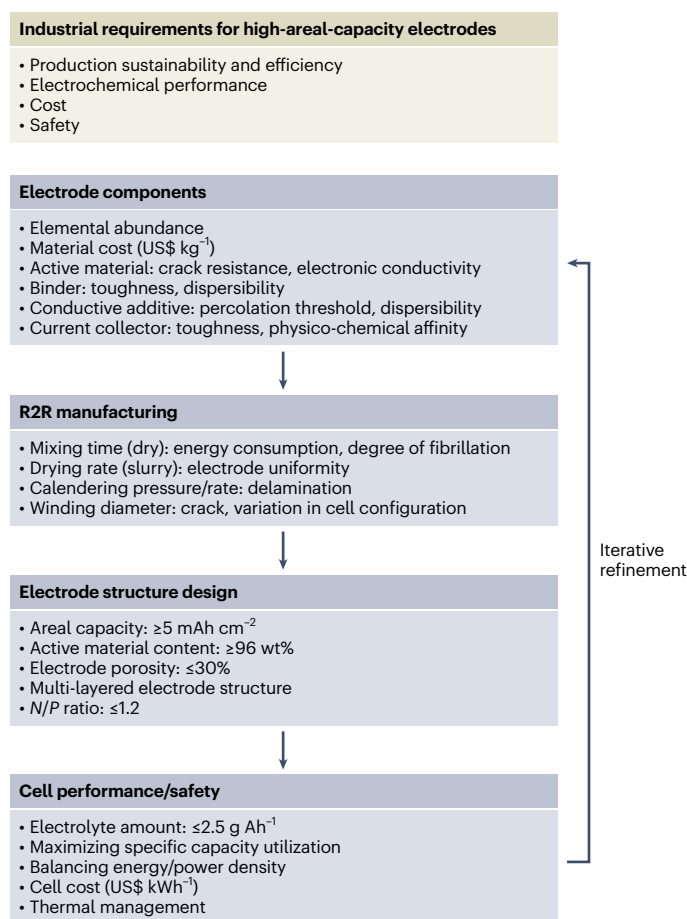


Fig. 4 | Development directions for scalable high-areal-capacity battery electrodes. Outline of comprehensive strategies for upscaling high-areal-capacity electrodes, focusing on electrode components, R2R manufacturing, electrode structure design and cell performance/safety. Critical parameters at each stage are identified by infusing the industrial requirements for production sustainability, efficiency, electrochemical performance, cost and safety. This mechanistic framework advances TRLs, providing a structured pathway to bridge the gap between laboratory-scale innovations and industrially scalable applications.

enhance solid-state diffusion inside the particles, whereas larger active materials in the bottom layer reduce tortuosity and mitigate the ion concentration gradient in electrodes, enabling high rate capabilities⁹⁶. Achieving commercial-level densities (above 3.2 g cm⁻³ for cathodes) in high-areal-capacity electrodes is challenging⁴ because low porosity impedes ion transport in thick electrodes. Innovative strategies include increasing the cation transference number of electrolytes and reducing the tortuosity inside electrodes. Another approach is the magnetic alignment of electrode materials⁹⁷, which has been implemented in the production line for fast-charging graphite anodes.

Owing to the increased power density challenge of high-areal-capacity electrodes, N/P ratios should be carefully adjusted to match the intended current rates. Although mild N/P ratios enhance the cycle life, they compromise the cell energy density. Therefore, a lower N/P ratio (<1.2) is preferred while addressing the aforementioned trade-off issue. In high-areal-capacity electrodes, increased thickness impedes electrolyte infiltration in the through-thickness direction⁹⁸, and is exacerbated at practical E/C ratios (<2.5 g Ah⁻¹). Given that electrode widths now extend up to 600 mm in EV batteries³², uniform wetting becomes challenging. This raises concerns about the industrial feasibility of high-concentration electrolytes with high viscosity, in addition to their high cost. Strategies such as electrolyte design,

engineering electrolyte injection and integrating electrocapillary ageing processes need to be investigated under such practical conditions⁹⁹. The deactivation of active materials paradoxically reduces the specific cell energies despite an increased M_{AM} , which is particularly pronounced at high current rates. Beyond merely increasing M_{AM} , longitudinal resistances must be managed to fully utilize them. Managing heat dissipation in cells with high-areal-capacity electrodes should also occur with this, as such electrodes inherently generate more heat than low-areal-capacity electrodes. Our BatPaC model⁹ calculations showed a 50% reduction in the in-plane thermal conductivity of the cell as $M_{AM,cat}$ is increased from 15 to 35 mg cm⁻². This reduction is attributed to the decreased number of thermally conductive current collectors incorporated in the cell, potentially exacerbating in-plane heat accumulation. By comparison, through-plane thermal conductivity increases 1.2-fold due to the reduced cell thickness. Effective thermal management requires the mechanistic engineering of cells and packs, in addition to the material-based approaches. Finally, establishing a feedback loop, as described in Fig. 4, is necessary to bridge laboratory-scale innovations with industrial scalability.

High-areal-capacity electrodes in post-LIB systems

Cell-level design extends beyond specific chemistries and can be universally applied to emerging batteries. Current research is exploring the potential of high-areal-capacity electrodes in various systems, including Na-ion batteries (NIBs), LMBs and all-solid-state batteries (ASSBs). However, an understanding of their operating mechanisms under high-capacity conditions remains in its infancy.

The natural abundance of Na makes NIBs promising for next-generation grid-scale energy storage systems, aiming to reduce market's reliance on LIBs. Although the cathode chemistry of NIBs is similar to that of LIBs, hard carbon anodes in NIBs exhibit higher surface areas (with nano-sized pores) and lower densities than graphite anodes. Further investigation is required into electrolyte infiltration and extended ion-diffusion pathways in thick, hard carbon anodes.

Metal anodes (for example, Li, Na and Zn) offer higher capacities than intercalation-based anodes. However, their plating/stripping behaviour under high-current/high-capacity conditions shows limited cycling stability due to dendrite formation. A further understanding of nucleation, growth and stripping behaviour under such challenging conditions is required.

Recent advances in ASSBs have shown a promising performance with high-areal-capacity electrodes in pouch-cell configurations¹⁰⁰. The use of cathodes with solid electrolytes enables ionic percolation without necessitating external electrolyte infiltration. The typical operating conditions for ASSBs, such as high pressures and elevated temperatures, are expected to suppress dendrite formation and improve the cyclability of high-areal-capacity metal cells. Whereas these conditions facilitate ASSBs in high-capacity applications, the restricted manufacturing environments (for example, argon atmospheres) and the compatibility of the sintering process with R2R manufacturing pose formidable challenges.

Whereas high-areal-capacity electrodes offer substantial advantages in specific energy and cost efficiency, they also present trade-offs that must be carefully managed. It is imperative to avoid the indiscriminate pursuit of high-areal-capacity without addressing the associated practical challenges. Future research should focus on achieving an optimal balance between performance and manufacturability, considering the overall cell-level factors.

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Author contributions

J.-H.K., N.-Y.K. and S.-Y.L. conceived the project. J.-H.K., N.-Y.K. and M.-S.P. investigated the industrial electrode manufacturing processes. J.-H.K. performed the specific energy and cost analyses of the cells. Z.J. conducted the theoretical calculations of cell performance. J.-H.K., K.-D.K. and S.-J.L. conducted the literature studies on slurry-cast electrode processing. J.-H.K., N.-Y.K., J.-H.P. and S.-S.C. carried out the literature studies on dry-coated electrode processing. J.-H.K., Z.J. and Y.-K.H. performed the literature studies on alternative electrode-processing methods. J.-Y.K., G.Y. and S.-Y.L. supervised the overall project. All authors contributed to the writing and revision of the manuscript.

Competing interests

The authors declare no competing interests.

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